

CASE STUDY

Digital Quality Platform — Tamil Nadu Automotive Cluster

Case Study | Tamil Nadu, South India | 2023

Executive Summary

We designed and deployed a shared digital platform for a consortium of five Tier-2 auto component manufacturers in the Tamil Nadu automotive cluster. The cloud-based solution integrated quality management, supplier tracking, and logistics visibility across 12 plants, delivering a 25% reduction in customer-facing defects and measurably improved on-time delivery.

25%	18pp	12	5
Defect Reduction	OTD Improvement	Plants Covered	Cluster Partners

Client Context

The Tamil Nadu automotive cluster supplies critical sub-assemblies to three large OEMs in Chennai and Pune. OEM quality audits had flagged high inter-plant variability and inadequate digital traceability. The five-member consortium needed a shared platform that would be cost-effective for mid-size suppliers while meeting OEM data-sharing requirements.

Scope of Work

- Design of a multi-tenant cloud architecture (Azure) supporting 5 independent company tenants
- Quality Management System (QMS) module: inspection plans, non-conformance workflow, CAPA tracking
- Supply chain visibility layer: real-time shipment tracking and inbound material quality gates
- OEM portal: read-only view for OEM quality teams to monitor supplier performance KPIs
- Mobile QC app for shop-floor inspection entry (works offline in low-connectivity plants)
- Data migration from 5 legacy ERP/QMS systems and paper-based processes

South India Manufacturing Expertise

Our South India practice has deep familiarity with the Tamil Nadu automotive ecosystem, including understanding of the supplier-OEM relationships, local regulatory requirements, and the specific challenges of mid-market manufacturers operating with lean IT teams. This contextual knowledge was critical in designing a solution with low operational overhead.

Key Outcomes

- 25% reduction in customer-reported defects across the consortium within 12 months
- On-time delivery improved by 18 percentage points (from 74% to 92%)
- All 5 OEM quality audits passed with 'green' rating post-implementation

- Shared platform economics reduced per-company digital investment by ~60% vs individual systems
- Programme recognised by CII Tamil Nadu as a model for SME cluster digitalisation